YEAR 12 MATHEMATICS EXTENSION 1

Chapter 13 Differential Equations

Lesson and Homework Pack

Revision	Exponential growth, logs, integration by substitution	Advanced · Chapter 12
Lesson 13A	Differential equations and their solutions	Past papers: 16 marks · 30 min
Lesson 13B	Slope fields	Past papers: 14 marks · 25 min
Lesson 13C	Separable differential equations	Past papers: 19 marks · 30 min
Lesson 13D	y′ = g(y) and the logistic equation	Past papers: 21 marks · 30 min
Lesson 13E	Applications of differential equations	Past papers: 22 marks · 35 min
Review	Chapter 13 review exercise	Mixed practice

Each lesson runs in the same order: **concept** → **tutor demo** → **your attempt** → **worked example** to keep for homework. Then do the **textbook homework** and finish with the **past-paper set** under exam conditions. In this chapter $\log x$ means $\log_e x$. Answers and full past-paper solutions are in the separate Homework Answers booklet.

CHAPTER REVISION

What this chapter builds on

Integration (Advanced and Chapter 12), exponential and logarithmic functions, and implicit differentiation with the chain rule.

SUMMARY · keep this page open while you work

LOGARITHMS AND EXPONENTIALS

$$\int \frac{1}{x}\,dx = \log|x| + C \qquad \int \frac{f'(x)}{f(x)}\,dx = \log|f(x)| + C \qquad \int e^{ax+b}\,dx = \tfrac{1}{a}e^{ax+b} + C$$

$$\log a - \log b = \log\frac{a}{b} \qquad e^{\log x} = x \qquad \log|y| = u \iff |y| = e^u$$

INVERSE TRIGONOMETRIC INTEGRALS (CHAPTER 12)

$$\int \frac{1}{a^2+x^2}\,dx = \frac{1}{a}\tan^{-1}\frac{x}{a} + C \qquad \int \frac{1}{\sqrt{a^2-x^2}}\,dx = \sin^{-1}\frac{x}{a} + C$$

DIFFERENTIATING EXPRESSIONS INVOLVING Y

By the chain rule, $\frac{d}{dx}\big(f(y)\big) = f'(y)\,\frac{dy}{dx}$. For example $\frac{d}{dx}(y^2) = 2yy'$ and, by the product rule, $\frac{d}{dx}(xy) = y + xy'$.

WARM-UP · answers are in the answers booklet

1 Find:

a $\displaystyle\int \frac{3}{x}\,dx$ b $\displaystyle\int \frac{2x}{x^2+1}\,dx$ c $\displaystyle\int e^{-2x}\,dx$

2 Solve for y: a $\log|y| = 3x + 2$ b $\displaystyle\log\left|\frac{y}{1-y}\right| = x$, given $0 < y < 1$.

3 Show that $\displaystyle\frac{1}{x(2-x)} = \frac{1}{2}\left(\frac{1}{x} + \frac{1}{2-x}\right)$.

4 Differentiate $x^2 + y^2 = 16$ with respect to x, and make $\frac{dy}{dx}$ the subject.

5 A population grows according to $P = 500e^{0.2t}$. Find P when $t = 5$, and the time when the population reaches 2000.

LESSON 13A

Differential equations and their solutions

What a differential equation (DE) is, its order, testing solutions by substitution, families of solutions, initial value problems, relations as solutions, and first-order linear DEs.

CONCEPT 1 · differential equations and their order

DEFINITIONS

- A **differential equation** (DE) is an equation that involves a derivative. For example $y' = -7y$, $y' = 1 + y^2$, $yy' + x = 0$, $y'' + 49y = 0$.
- The **order** of a DE is the order of the highest derivative that occurs in it. $y'' + 49y = 0$ and $x^2y'' - xy' + y = 0$ have order 2; the others above have order 1.
- Its solutions are not numbers but **functions** (or, more generally, relations in x and y). To **test** whether a function is a solution, substitute it into the DE.
- An indefinite integral $\int f(x)\,dx$ is equivalent to the DE $y' = f(x)$.

TUTOR DEMO Order, and testing a solution

a State whether each is a DE, and if so its order: (i) $y''' + y = x$ (ii) $x^2 + y^2 = 4$ (iii) $y' = \sin x$.
b Test whether each function is a solution of $y' = 3y$: (i) $y = 5e^{3x}$ (ii) $y = e^x + 3$.

YOUR TURN Same skills

a State the order of: (i) $y'' - 4y = 0$ (ii) $xy' + y = x^2$.
b Show that $y = \tan x$ is a solution of $y' = 1 + y^2$.

WORKED EXAMPLE 1 **Keep for homework**

Test whether each function is a solution of the differential equation $y' = -7y$: **a** $y = 20e^{7x}$ **b** $y = 20e^{-7x}$.

SOLUTION

a Differentiating, $y' = 140e^{7x}$. Substituting into $y' = -7y$:

$\text{LHS} = 140e^{7x} \qquad \text{RHS} = -7 \times 20e^{7x} = -140e^{7x}$

LHS $\neq$ RHS, so the function is not a solution.

b Differentiating, $y' = -140e^{-7x}$:

$\text{LHS} = -140e^{-7x} \qquad \text{RHS} = -7 \times 20e^{-7x}$ =

LHS = RHS, so the function is a solution.

The DE involves only algebraic operations, yet its solution is exponential. This is typical of DEs, just as $\int \frac{dx}{1+x^2} = \tan^{-1} x + C$.

CONCEPT 2 · a family of solutions, and particular solutions

GENERAL AND PARTICULAR SOLUTIONS

- The general solution of a DE normally involves one or more **arbitrary constants**, giving a **family** of solutions. The number of arbitrary constants is normally equal to the **order** of the DE.
- These constants can be evaluated to give **particular solutions** if one or more **initial conditions** (boundary conditions) are known. The problem of finding a particular solution is called an **initial value problem** (IVP).
- For a higher-order DE the initial conditions may also involve values of y', and usually give simultaneous equations.
- A **solution curve** (integral curve) may be just one of the connected branches of a function.

TUTOR DEMO **One arbitrary constant**

a Show that $y = Ae^{2x} - 1$ is a solution of $y' = 2y + 2$ for all values of A. **b** Find the particular solution with $y(0) = 3$.

YOUR TURN **Same method**

a Show that $y = Ce^{-x} + x - 1$ is a solution of $y' = x - y$ for all values of C. **b** Find the particular solution through $(0, 2)$.

TUTOR DEMO **Two arbitrary constants**

Show that $y = A\cos 3x + B\sin 3x$ satisfies $y'' = -9y$ for all A and B, and find the solution for which $y(0) = 2$ and $y'(0) = 6$.

YOUR TURN **Integrating twice**

Solve the IVP $y'' = 6x$, with $y(0) = 1$ and $y'(0) = -2$.

WORKED EXAMPLE 2 **Keep for homework**

a Show that $y = Ae^x - x - 1$ is a solution of $y' = x + y$, for all values of A.
b Find the particular solutions (i) through the origin, (ii) with $y = -2$ when $x = 0$, (iii) through $(0, -1)$.

SOLUTION

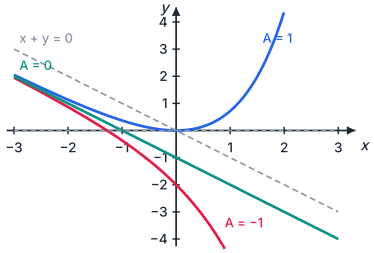
a Substituting $y = Ae^x - x - 1$:

$$\text{LHS} = y' = Ae^x - 1 \qquad \text{RHS} = x + (Ae^x - x - 1) = Ae^x - 1$$

Hence $y = Ae^x - x - 1$ is a solution for all values of A.

b (i) $0 = A - 0 - 1$, so $A = 1$: $y = e^x - x - 1$.
(ii) $-2 = A - 0 - 1$, so $A = -1$: $y = -e^x - x - 1$.
(iii) $-1 = A - 0 - 1$, so $A = 0$: $y = -x - 1$.

Substituting $y' = 0$ into the DE gives the line $x + y = 0$: a solution curve has a stationary point only where it crosses this line. Only the curve in (i) does (a minimum at the origin).



WORKED EXAMPLE 3 **Keep for homework**

It is known that $y = A\sin 7x + B\cos 7x$ is a solution of $y'' = -49y$ for all A and B. Find the solution for which $y(0) = 1$ and $y'(0) = 14$.

SOLUTION

$$y = A\sin 7x + B\cos 7x, \qquad y' = 7A\cos 7x - 7B\sin 7x$$

When $x = 0$, $y = 1$, so $1 = 0 + B$. When $x = 0$, $y' = 14$, so $14 = 7A + 0$. Hence $A = 2$, $B = 1$, and $y = 2\sin 7x + \cos 7x$.

CONCEPT 3 · relations as solutions, and linear DEs

RELATIONS CAN BE SOLUTIONS

Relations that are not functions can be solutions of DEs, with the usual arbitrary constants and families of curves. Forming the DE from such a relation uses the chain rule, for example $\frac{d}{dx}(y^2) = 2yy'$ and $\frac{d}{dx}(xy) = y + xy'$.

FIRST-ORDER LINEAR DES

- A first-order DE is **linear** if it can be put into the form $y' + f(x)\,y = g(x)$.
- An indefinite integral is equivalent to a first-order linear DE with $f(x) = 0$: $y' = g(x)$.
- An exponential growth DE $y' - ky = 0$ is first-order linear with $g(x) = 0$ and $f(x)$ constant.

For example, $y' = \frac{y}{x}$ is linear (it is $y' - \frac{1}{x}y = 0$), but $y' = \frac{x}{y}$ is not (it becomes $yy' = x$).

TUTOR DEMO **A relation as a solution**

Differentiate $x^2 + y^2 = a^2$ with respect to x to show that every circle in this family satisfies $y' = -\dfrac{x}{y}$.

What geometric fact does this express?

YOUR TURN **Same method**

Show that the hyperbola $xy = a^2$ satisfies $y' = -\dfrac{y}{x}$. Is this DE linear?

TEXTBOOK HOMEWORK · Exercise 13A

FOUNDATION

1 For each DE, state: (i) its order, (ii) whether it is linear (first-order DEs only), (iii) how many arbitrary constants will appear in the general solution.

a $y' - y = x$ b $y'y = 3x$ c $y'' + 4y' - y = \sin x$

d $y' + y\cos x = e^x$ e $y'' - \frac{1}{2}(y')^2 = 0$ f $y' + y^2 = 1$

g $y' + xy = 0$ h $xy'' + y' = x^2$ i $y'' - xy' + e^x y = 0$

2 Show by substitution that the given function is a particular solution of the DE.

a $y = 5x^3;\quad xy' - 3y = 0$ b $y = x^2 - 1;\quad xy' - 2y = 2$

c $y = 3e^{-x};\quad y' + y = 0$ d $y = \sqrt{x^2 + 4};\quad y'y = x$

3 Find the general solution of each DE by integration.

a $y' = 2x - 3$ b $y' = 12e^{-2x} + 4$ c $y' = \sec^2 x$

d $y' = 6\cos 2x + 9\sin 3x$ e $y' = \sqrt{1 + 5x}$ f $y' = 4x\cos x^2$

DEVELOPMENT

4 Show by substitution that the given function with arbitrary constant C is the general solution of the DE.

a $y = Ce^x - x - 1; \quad y' = x + y$

b $y = Cxe^{-x}; \quad xy' = y(1 - x)$

c $y = \sin(x + C); \quad (y')^2 = 1 - y^2$

d $y = \dfrac{C}{x} + 2; \quad \dfrac{dy}{dx} = \dfrac{2 - y}{x}$

5 Verify that the given function is a particular solution of the DE.

a $x^2y'' - 2xy' + 2y = 6; \quad y = x^2 - 2x + 3$

b $y'' - 6y' + 5y = 0; \quad y = 2e^x + e^{5x}$

c $y'' + 4y' + 5y = 0; \quad y = e^{-2x}\sin x$

d $y'' + \pi^2 y = 0; \quad y = \cos \pi x - 3 \sin \pi x$

6 Solve these second-order DEs by integrating twice.

a $y'' = 2$

b $y'' = \cos 2x$

c $y'' = e^{\frac{1}{2}x}$

d $y'' = \sec^2 x$

7 a Show that $y = e^{-x}$ and $y = e^{3x}$ are each solutions of $y'' - 2y' - 3y = 0$.

b Now show that $y = Ae^{-x} + Be^{3x}$ is also a solution of this equation for any values of the constants A and B.

8 Solve these initial value problems. In each case, use integration to find the general solution, then use the initial condition to evaluate the constant.

a $y' = 1;\ y(2) = 1$

b $y' = 2x - 3;\ y(0) = 2$

c $y' = 3x^2 + 6x - 9;\ y(1) = 2$

d $y' = \sin x;\ y(\pi) = 3$

e $y' = 6e^{2x};\ y(0) = 0$

f $y' = 3\sqrt{x} - 2;\ y(4) = 7$

9 Consider the IVP $y' = \dfrac{1}{x(1-x)}$, where $y\left(\frac{1}{2}\right) = 0$.

a Show that $\dfrac{1}{x(1-x)} = \dfrac{1}{x} + \dfrac{1}{1-x}$.

b Find the general solution of the DE.

c Hence solve the IVP.

10 a (i) Differentiate both sides of $x^2 + y^2 = 9$ with respect to x. (ii) Hence show that $\dfrac{dy}{dx} = -\dfrac{x}{y}$ at each point on the circle. (iii) Where is this DE undefined, and why is this expected?

b Likewise, find $\dfrac{dy}{dx}$ for: (i) the parabola $y^2 = x + 4$ (ii) the hyperbola $xy = c^2$ (iii) the ellipse $9x^2 + 16y^2 = 144$ (iv) the hyperbola $x^2 - 4y^2 = 4$

11 a Show that $y = \sin x$ is a solution of $y'' + y = 0$.

b Find the value of the second derivative y'' when $y = 12$.

c Suppose that $y = f(x)$ is a solution of this DE. What does the equation say about $f''(x)$? Answer in terms of a transformation of $f(x)$.

12 Use integration and the initial conditions to find the solution of $y'' = \sec^2 x$ with $y(0) = 1$ and $y'(0) = 1$.

ENRICHMENT

13 For each DE, use substitution to find the values of λ, if any, that make $y = Ae^{\lambda x}$ a solution for all values of A.

a $y'' - 4y' + 3y = 0$ b $y'' + 2y' + y = 0$ c $y'' - 3y' + 4y = 0$

PAST-PAPER HOMEWORK · mini exam

Suggested time: 30 minutes **16 marks** Calculator allowed · show working · hints are below each question if you get stuck

PP 1 **Abbotsleigh, 2023, Higher School Certificate Assessment 4 Trial Examination, Mathematics Extension 1, Q7** 1 mark

What are the values of p for which $y = e^{px}$ satisfies the differential equation

$$\frac{d^2y}{dx^2} - \frac{dy}{dx} - 12y = 0\,?$$

(A) $p = -4,\ p = 3$ (B) $p = -3,\ p = 4$ (C) $p = \pm 3$ (D) $p = \pm 4$

Hint: substitute $y = e^{px}$, $y' = pe^{px}$, $y'' = p^2e^{px}$ and divide by e^{px}.

PP 2 **Cherrybrook Technology High School, 2020, Year 12 AP4 Mathematics Extension 1 (Trial), Q8** 1 mark

The equation $y = e^{ax}$ satisfies the differential equation $\frac{d^2y}{dx^2} + \frac{dy}{dx} - 6y = 0$. What are the possible values of a?

A. $a = -2$ or $a = 3$ B. $a = -1$ or $a = 6$ C. $a = 2$ or $a = -3$ D. $a = 1$ or $a = -6$

Hint: same method as PP 1.

PP 3 **Sydney Technical High School, 2020, Trial Higher School Certificate Examination (Sample 1), Mathematics Extension 1, Q11(c)** 2 marks

Show that $y = x \ln x$ satisfies the differential equation $x^2y'' - xy' + y = 0$.

Hint: find y' by the product rule, then y'', and substitute all three.

PP 4 **Gosford High School, 2022, Trial Higher School Certificate Examination, Mathematics Extension 1, Q11(a)** 2 marks

Show that $y = Ae^{3x} + Be^{-2x}$ satisfies the differential equation $y'' - y' - 6y = 0$ for any real values of A and B.

Hint: collect the e^{3x} terms and the e^{-2x} terms separately.

PP 5 **Sydney Boys High School, 2020, Year 12 Term 3 Trial HSC, Mathematics Extension 1, Q13(a)** 5 marks

A model for the population, K, of kangaroos is

$$K = \frac{21\,000}{7 + 3e^{-\frac{t}{3}}},$$

where t is the time in years from today.

(i) Show that K satisfies the differential equation $\dfrac{dK}{dt} = \dfrac{1}{3}\left(1 - \dfrac{K}{3000}\right)K$. 2

(ii) What is the population today? 1

(iii) What does the model predict that the eventual population will be? 1

(iv) What is the annual percentage rate of growth today? 1

Hint: in (i) work out $1 - \dfrac{K}{3000}$ as a single fraction first. In (iv), the percentage rate is $\dfrac{1}{K}\dfrac{dK}{dt} \times 100\%$.

PP 6 **Blacktown Boys' High School, 2020, HSC Trial Examination, Mathematics Extension 1, Q13(b)** 5 marks

Metal fatigue is a phenomenon where a piece of steel will fail when repeatedly subjected to force F. The endurance limit is the force below which the steel will not break even if subjected to an infinite number of applications of that force. Let the number of applications be n. The force and the number of applications are related by the differential equation $\frac{dF}{dn} = -k(F - F_0)$, where k and F_0 are constants.

(i) Show that $F = 275e^{-k(n-1)} + F_0$ is a solution to the differential equation. 1

(ii) If $F = 350$ when $n = 1$, find the value of F_0. 1

(iii) Find the exact value of k if $F = 80$ when $n = 200$. 2

(iv) Find the endurance limit. 1

Hint: the endurance limit is what F approaches as $n \to \infty$.

LESSON 13B

Slope fields

Draw and read slope fields for DEs of the form $y' = G(x, y)$, sketch solution curves, use isoclines, and recognise constant (equilibrium) solutions.

CONCEPT 1 · drawing a slope field

SLOPE FIELDS (GRADIENT FIELDS, DIRECTION FIELDS)

A slope field can only be drawn for a first-order DE that can be written with y' as the subject, $y' = G(x, y)$. At each point on a solution curve the tangent has gradient $G(x, y)$, so we draw a short **line element** with that gradient.

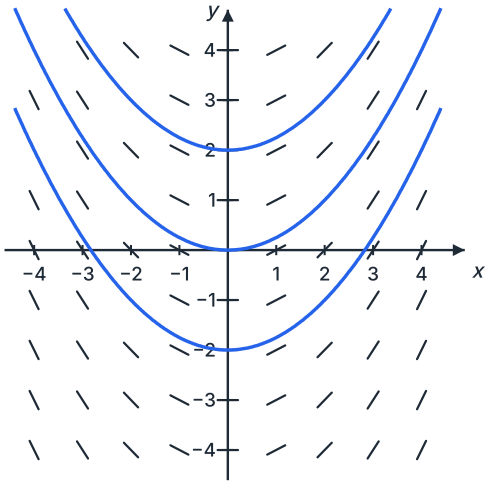
- Choose a suitable grid of points in the plane.
- Draw up a table showing the gradients at these grid points.
- Display these gradients by short tilted line elements of a fixed length, centred at the grid points.

Each line element shows only a gradient: it has no direction, and its length has no significance.

EXAMPLE: Y' = ½X

All the rows of the table of slopes are the same, because y is not involved. The slope field suggests a family of parabolas with the y-axis as axis of symmetry, all vertical translations of each other. Indeed, integrating gives $y = \frac{1}{4}x^2 + C$.

Every vertical line is an **isocline** (a curve through points where the gradient is the same). The y-axis is special: the gradient is zero there, so every solution curve has a stationary point where it crosses the y-axis.



TUTOR DEMO **Table, then field**

For $y' = x - y$: **a** find the gradients at the nine points with $x, y \in \{-1, 0, 1\}$; **b** draw these line elements; **c** identify the isoclines.

YOUR TURN **Same method**

For $y' = xy$: find the gradients at the nine points with $x, y \in \{-1, 0, 1\}$, draw the line elements, and describe where the gradient is zero.

CONCEPT 2 · sketching solution curves and reading a field

SKETCHING A SOLUTION CURVE (INTEGRAL CURVE)

- Place a pencil on the diagram and follow the gradients. Two solution curves can never cross (apart from strange singularities), because the gradient at each point can only have one value.
- This is not an accurate procedure. In particular, asymptotic behaviour may not be clear.
- If there is an initial condition, start there and draw in both directions. Otherwise draw about three representative curves.

THINGS TO LOOK FOR IN A SLOPE FIELD

- Where the slope is **zero** (horizontal line elements): solution curves have stationary points there.
- Where the line elements slope **upwards** (positive) and **downwards** (negative): this tells where curves increase and decrease.
- **Isoclines**, and any isocline whose line elements lie along the isocline itself: such a line is a solution curve, no other solution crosses it, and nearby curves often have it as an asymptote.
- **Vertical** line elements, where the DE has a division by zero.

These features are the quickest way to eliminate wrong options when matching a DE to a slope field.

WORKED EXAMPLE 1 **Keep for homework**

a Construct a slope field for $y'y + x = 0$. **b** What family of curves look as if they are solution curves? **c** Check by differentiation.

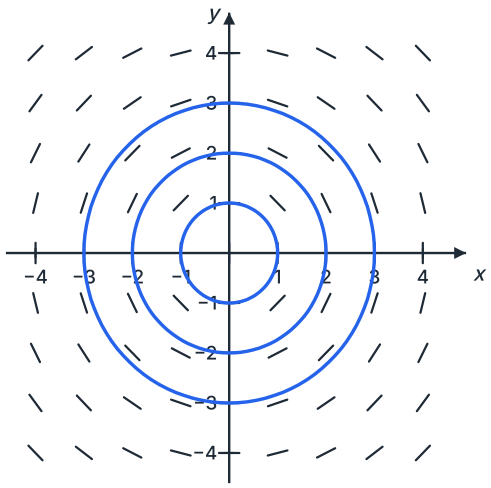
SOLUTION

a Solving for y' gives $y' = -\dfrac{x}{y}$, provided that $y \neq 0$. On the x-axis the line elements are drawn vertical, because there they arise from division of a non-zero number by zero. The origin has no line element at all.

The gradient is zero on the y-axis (excluding the origin), negative in quadrants 1 and 3 and positive in 2 and 4. The isoclines are all the lines through the origin.

b The solution curves look like circles with centre the origin.

c Testing $x^2 + y^2 = r^2$: differentiating, $2x + 2yy' = 0$, as required.



TUTOR DEMO **Match a DE to a field**

A slope field has zero gradient along the y-axis, positive slopes in quadrant 2 and negative slopes in quadrants 1 and 3, and the slopes become steeper away from the axes. Which of these could it be: A $y' = -xy$ B $y' = -x^2y$ C $y' = -xy^2$ D $y' = -x - y$?

CONCEPT 3 · isoclines that are solutions, and constant solutions

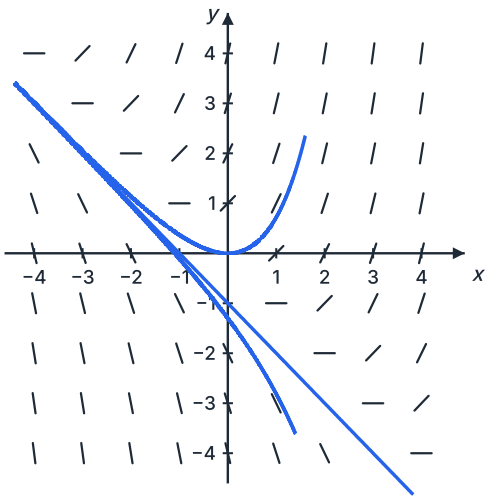
AN ISOCLINE THAT IS AN ASYMPTOTE

For $y' = x + y$, every line $x + y = c$ is an isocline (gradient c). The isocline $y = -x - 1$ is exceptional: its gradient -1 matches the line elements along it, so it is itself a solution. No other solution crosses it; the curves above it curl upwards, those below curl downwards, and all are asymptotic to it on the left.

CONSTANT SOLUTIONS ARE HORIZONTAL ASYMPTOTES

Any constant solution $y = k$ stands out on a slope field as a horizontal line of horizontal line elements. It is an isocline, and it divides the solutions into those above and those below. Usually many solution curves have it as a horizontal asymptote on the left or right.

A constant solution that is an asymptote on the **right** is a **stable equilibrium**; one that is an asymptote on the **left** is an **unstable equilibrium**.



WORKED EXAMPLE 2 **Keep for homework**

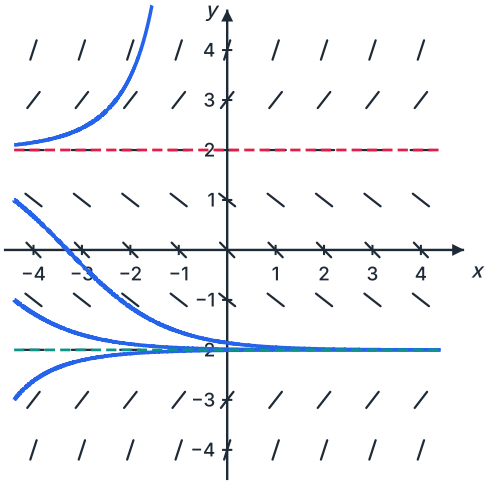
For the DE $y' = \frac{1}{4}(y-2)(y+2)$: **a** identify the constant solutions from the slope field; **b** show that they are solutions; **c** if the horizontal axis is time, describe the behaviour of solution curves near each constant solution.

SOLUTION

a The horizontal isoclines $y = 2$ and $y = -2$ stand out because both consist of points where the gradient is zero.

b Substituting $y = 2$ or $y = -2$ gives LHS $= y' = 0$ and RHS $= 0$.

c As time increases, the curves between and below the lines approach $y = -2$: this is a **stable** equilibrium. A curve starting just above $y = 2$ goes up without bound, and one just below goes down towards $y = -2$, so $y = 2$ is an **unstable** equilibrium.



YOUR TURN **Constant solutions**

For $y' = y(3-y)$, find the constant solutions, describe the sign of y' in each region they create, and state which is stable.

TEXTBOOK HOMEWORK · Exercise 13B

FOUNDATION

1 In each case, find the value of y' at the given point.

a $y' = 2x - 3$ at $(1, 1)$

b $y' = 2\cos x - 1$ at $(0, 0)$

c $y' = 4 - y^2$ at $(0, 1)$

d $y' = \dfrac{1}{1+y}$ at $(3, 1)$

e $y' = \dfrac{y}{x} + 1$ at $(-2, 1)$

f $y' = xy - x$ at $(1, -2)$

2 Answer these questions for the differential equation $y' = \frac{1}{2}x - 1$.

a Complete a table of values of y' for $x = -1, 0, 1, \ldots, 5$ and $y = -3, -2, \ldots, 3$.

b Draw a number plane with domain $[-1, 5]$ and range $[-3, 3]$, and through each grid point draw a short line element with the gradient in the table.

c Notice that the vertical line $x = 1$ is an isocline. Why is this expected from the table?

d The slope field indicates a positive gradient to the right of $x = 2$ and a negative gradient to the left of $x = 2$. What will be the concavity of a solution curve?

e Starting at the origin, draw an integral curve to the right and to the left. What type of curve might this be?

f Draw two more integral curves, starting at $(0, 2)$ and $(0, 3)$.

3 Consider the differential equation $y' + y = x$.

a Make y' the subject.

b Complete a table of values for $x = -1, -\frac{1}{2}, 0, \frac{1}{2}, 1, \frac{3}{2}, 2$ and $y = 1, \frac{1}{2}, 0, -\frac{1}{2}, -1, -\frac{3}{2}, -2$, and draw the slope field.

c Look carefully at the table for matching entries. Check that these agree with the isoclines in your graph.

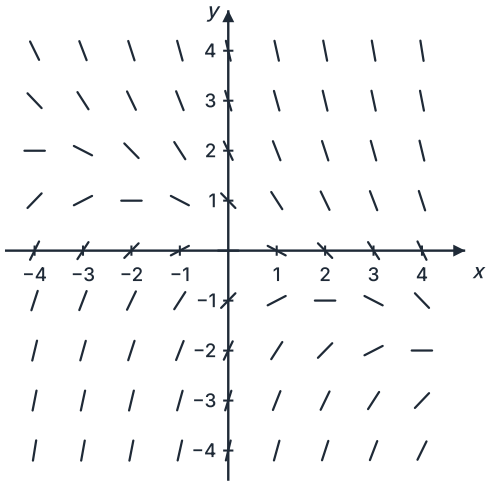
d What concavity would you expect for the solution curve passing through the origin? Draw this solution curve.

e Add solution curves through $(-1, -1)$ and $(1, -1)$.

f Which line do all your solution curves appear to have as an asymptote? Is this line a solution of the differential equation?

DEVELOPMENT

4 The slope field for the differential equation $y' = -\frac{1}{2}x - y$ is drawn below.



a Draw the solution curves through the points $(-2, 0)$ and $(2, 0)$.

b Look carefully at the slope of the line elements on the vertical line $x = 1$, from bottom to top. (i) Do the gradients increase or decrease? (ii) Do your solution curves converge or diverge as they cross $x = 1$, from left to right?

c Explain why the line $y = \frac{1}{2} - \frac{1}{2}x$ is an isocline.

d Show that the equation of this isocline is also a solution of the DE, then add the solution to your diagram.

e What do you notice about the isocline and the two solution curves you have drawn?

5 Consider the differential equation $y' = \dfrac{9 - y^2}{9}$.

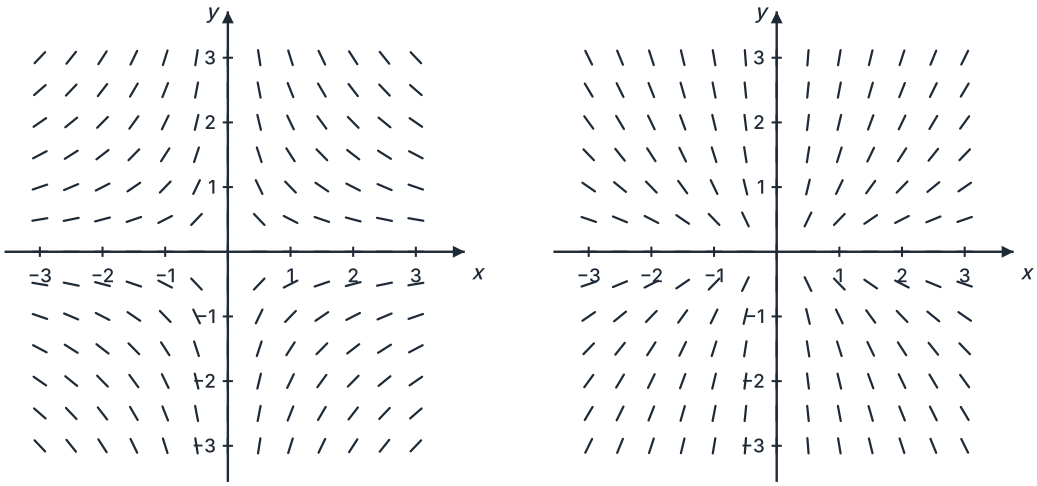
a Draw the slope field for this DE.

b What are the constant solutions for this DE?

c Are these constant solutions isoclines?

d Draw the solution curve through $(0, 0)$, and describe its shape.

6 The slope fields for $y' = -\dfrac{y}{x}$ and $y' = \dfrac{2y}{x}$ are shown.



a On the first field, add the solution curves through $(2, 2)$ and $(-2, -2)$, and show that the hyperbola $xy = 4$ is the solution of this IVP.

b On the second field, add the solution curve through $(2, 2)$, and show that the parabola $y = \frac{1}{2}x^2$ is the solution of this IVP.

c Show that $x^2 - y^2 = 4$ satisfies $y' = \dfrac{x}{y}$, and that $x^2 + 2y^2 = 4$ satisfies $y' = -\dfrac{x}{2y}$.

d Explain why the hyperbolas $xy = 4$ and $x^2 - y^2 = 4$ are perpendicular where they intersect.

ENRICHMENT

7 Prove the following three statements about isoclines.

a If $y' = f(x)$, then the isoclines $y' = c$ are the vertical lines.

b If $y' = g(y)$, then the isoclines $y' = c$ are the horizontal lines.

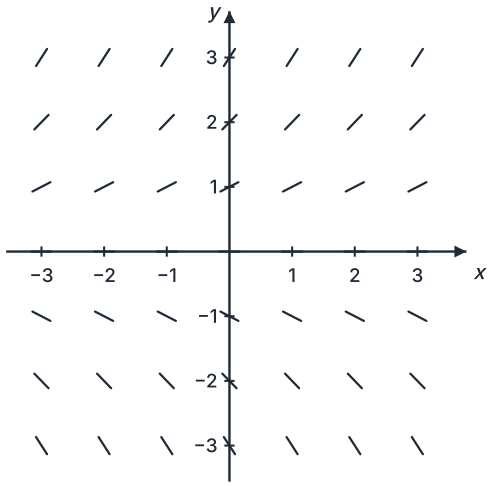
c If $y = cx + b$ is a solution of a first-order DE for a specific value of the constant c, then this line is also an isocline.

PAST-PAPER HOMEWORK · mini exam

Suggested time: 25 minutes **14 marks** Show working · hints are below each question if you get stuck

PP 1 **Blacktown Boys' High School, 2022, HSC Trial Examination, Mathematics Extension 1, Q2** 1 mark

Consider the slope field below. What is the possible differential equation for the slope field?

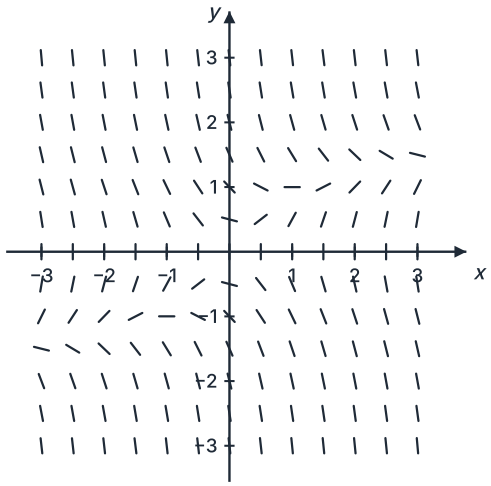


A. $\frac{dy}{dx} = -\frac{x}{2}$ B. $\frac{dy}{dx} = -\frac{y}{2}$ C. $\frac{dy}{dx} = \frac{x}{2}$ D. $\frac{dy}{dx} = \frac{y}{2}$

Hint: do the line elements change as you move left–right, or up–down?

PP 2 **Gosford High School, 2022, Trial Higher School Certificate Examination, Mathematics Extension 1, Q3** 1 mark

The slope field for a first order differential equation is shown. Which of the following best represents the differential equation shown in the slope field?

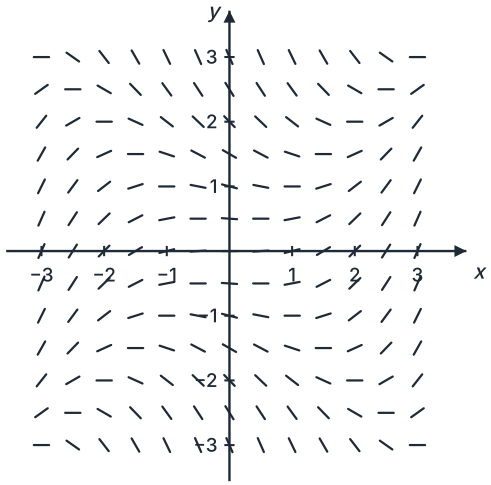


A. $\dfrac{dy}{dx} = \dfrac{x}{y} - y^2$ B. $\dfrac{dy}{dx} = \dfrac{x}{y} + y^2$ C. $\dfrac{dy}{dx} = -\dfrac{x}{y} - y^2$ D. $\dfrac{dy}{dx} = -\dfrac{x}{y} + y^2$

Hint: test the gradient at points such as $(1, 1)$ and $(1, -1)$, and compare with the line elements there.

PP 3 **Normanhurst Boys' High School, 2023, Year 12 Mathematics Extension 1 Course Assessment Task 4 (Trial Examination), Q7** 1 mark

Which of the following differential equations best corresponds to the slope field shown below?

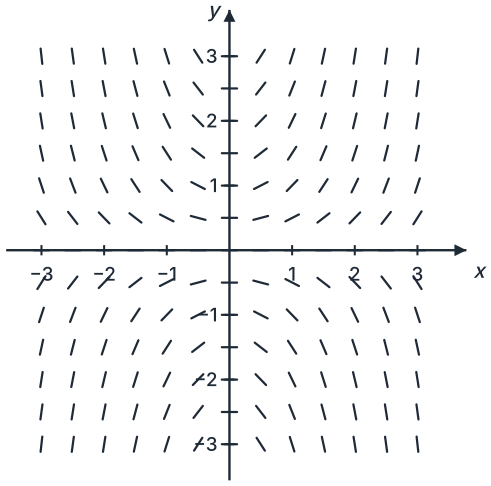


(A) $y' = \frac{1}{4}(-x^2 + y^2)$
(B) $y' = -\frac{1}{4}(x^2 + y^2)$
(C) $y' = \frac{1}{4}(x^2 - y^2)$
(D) $y' = \frac{1}{4}(x^2 + y^2)$

Hint: the slopes are not all one sign. Check the sign on the y-axis above the origin.

PP 4 **Kinross Wolaroi School, 2023, Year 12 Mathematics Extension 1 Trial HSC Examination, Q7** 1 mark

Consider the slope field below. Which of the following equations could have generated the above slope field?

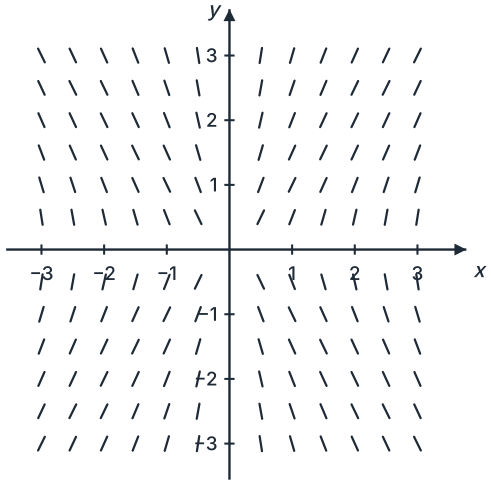


(A) $\frac{dy}{dx} = xy$ (B) $\frac{dy}{dx} = x^2y$ (C) $\frac{dy}{dx} = xy^2$ (D) $\frac{dy}{dx} = x^2y^2$

Hint: look at the sign of the slope in each of the four quadrants.

PP 5 **Abbotsleigh, 2024, Higher School Certificate Assessment 4 Trial Examination, Mathematics Extension 1, Q9** 1 mark

The diagram shows the direction field of a differential equation. The differential equation is undefined along both axes. Which of the following differential equations is best represented by the direction field above?

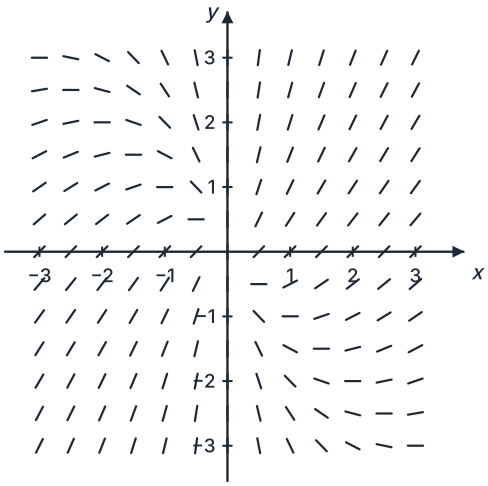


A. $\frac{dy}{dx} = \frac{xy}{x+y}$ B. $\frac{dy}{dx} = \frac{x}{y} + \frac{y}{x}$ C. $\frac{dy}{dx} = \frac{x+y}{xy}$ D. $\frac{dy}{dx} = \frac{x}{y} - \frac{y}{x}$

Hint: which options are undefined on both axes? Then look for lines of zero gradient.

PP 6 **Ascham School, 2024, Year 12 Trial Examination, Mathematics Extension 1, Q7** 1 mark

The diagram below shows the direction field for a differential equation. Which of the following differential equations could be represented by this direction field?

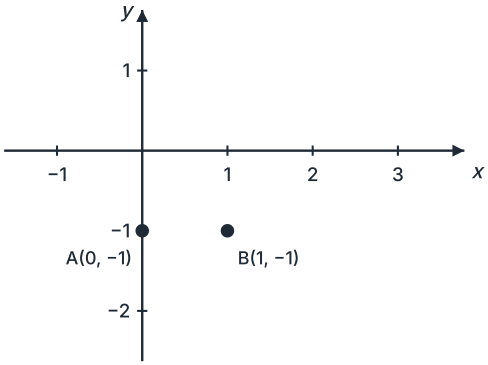


A. $\dfrac{dy}{dx} = \dfrac{y-x}{y}$ B. $\dfrac{dy}{dx} = \dfrac{y+x}{y}$ C. $\dfrac{dy}{dx} = \dfrac{x-y}{x}$ D. $\dfrac{dy}{dx} = \dfrac{x+y}{x}$

Hint: where are the line elements vertical? Then find where they are horizontal.

PP 7 **Blacktown Boys' High School, 2023, Year 12 Trial Examination, Mathematics Extension 1, Q12(a)** 2 marks

A direction field is to be drawn for the differential equation $\dfrac{dy}{dx} = \dfrac{2x-y}{y^2+x^2}$. Copy the diagram below, then clearly draw the correct slopes of the direction field at the points A and B.



Hint: calculate the gradient at each point first, then draw a short line element with that gradient.

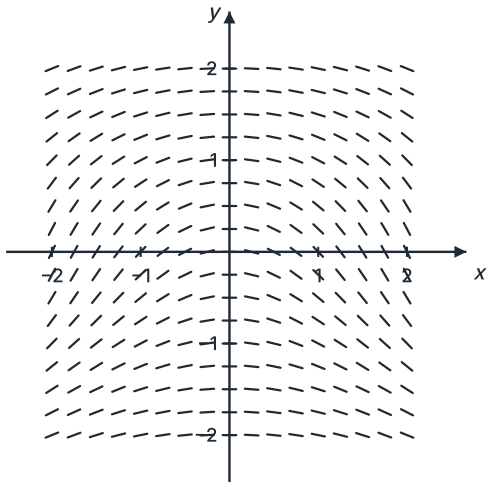
PP 8 **Knox Grammar School, 2023, Higher School Certificate Trial Examination, Mathematics Extension 1, Q12(a)** 2 marks

Consider the differential equation $\dfrac{dy}{dx} = \dfrac{2x+y}{x^3-y^3}$. Draw the correct slopes of the direction field at the points $A(0,-2)$, $B(-1,2)$ and $C(1,1)$.

Hint: one of the three points gives a zero denominator. What does that mean for the line element?

PP 9 **Sydney Technical High School, 2020, Trial Higher School Certificate Examination (Sample 3), Mathematics Extension 1, Q14(d)** 4 marks

A slope field representing the differential equation $\dfrac{dy}{dx} = \dfrac{-x}{1+y^2}$ is shown below.



(i) Sketch the solution curve of the differential equation corresponding to the condition $y(-1) = 1$ on the slope field above and, hence, estimate the positive value of x when $y = 0$. Give your answer correct to one decimal place. 2

(ii) Solve the differential equation $\dfrac{dy}{dx} = \dfrac{-x}{1+y^2}$ with the condition $y(-1) = 1$. Express your answer in the form $ay^3 + by + cx^2 + d = 0$, where a, b, c and d are integers. 2

Hint: in (ii) separate the variables: $(1+y^2)\,dy = -x\,dx$. Lesson 13C shows this method.

LESSON 13C

Separable differential equations

Solve DEs of the form $y' = f(x)\,g(y)$ by separating the variables and integrating, always looking first for constant solutions, and dealing carefully with absolute values.

CONCEPT 1 · separating the variables

SEPARABLE DIFFERENTIAL EQUATIONS

A first-order DE is **separable** if it can be put into the form $y' = f(x)\,g(y)$.

- The general case is $y' = f(x)\,g(y)$, where neither function is a constant.
- The case $y' = f(x)$, where $g(y) = 1$, is equivalent to an indefinite integral.
- The case $y' = g(y)$, where $f(x) = 1$, is Lesson 13D.

To solve the general case, separate the dx and dy and integrate both sides:

$$\frac{1}{g(y)}\,dy = f(x)\,dx \qquad \Longrightarrow \qquad \int \frac{1}{g(y)}\,dy = \int f(x)\,dx.$$

Justification: dividing $\dfrac{dy}{dx} = f(x)g(y)$ by $g(y)$ and integrating with respect to x gives $\displaystyle\int \frac{1}{g(y)}\frac{dy}{dx}\,dx = \int f(x)\,dx$; the chain rule then lets us cancel dx, as with substitution in Chapter 12.

TUTOR DEMO **Separate, integrate, solve for y**

a Solve $y' = 3x^2e^{-y}$. **b** Find the solution curve through $(0, 0)$.

YOUR TURN **A relation as the answer**

a Solve $y' = \dfrac{2x}{y}$. **b** Find the solution passing through $(1, 3)$.

WORKED EXAMPLE 1 **Keep for homework**

a Solve $y' = -2xe^{y}$, and find the solution curve through $(0, 0)$. **b** Solve $y' = \dfrac{x}{y}$, and find the solution passing through $P(4, 5)$.

SOLUTION

a Arrange the equation as $-e^{-y}\,dy = 2x\,dx$ and integrate:

$$\int -e^{-y}\,dy = \int 2x\,dx$$
$$e^{-y} = x^2 + C$$
$$y = -\log(x^2 + C)$$

Substituting $(0, 0)$: $0 = -\log C$, so $C = 1$ and $y = -\log(x^2 + 1)$.

b Separating, $y\,dy = x\,dx$:

$$\int y\,dy = \int x\,dx$$
$$\tfrac{1}{2}y^2 = \tfrac{1}{2}x^2 + C$$
$$y^2 - x^2 = D \quad (D = 2C)$$

Substituting $P(4, 5)$: $25 - 16 = D$, so the solution is $y^2 - x^2 = 9$.

CONCEPT 2 · look for constant solutions first

CONSTANT SOLUTIONS

- Always check first whether any constant functions $y = k$ are solutions of the DE: substitute $y' = 0$.
- This matters because when the DE is solved by separation, constant solutions are often lost through a division by zero.
- They are horizontal isoclines of horizontal line elements, and are usually horizontal asymptotes of nearby solution curves.

TUTOR DEMO **A lost solution**

a Solve $y' = 2xy^2$. **b** Find the particular solution given (i) $y(1) = -1$, (ii) $y(0) = 0$.

YOUR TURN **Same method**

Solve $y' = \dfrac{y^2}{x^2}$, and find the solutions with (i) $y(1) = \frac{1}{2}$, (ii) $y(2) = 0$.

WORKED EXAMPLE 2 **Keep for homework**

a Solve $y' = -xy^2$. **b** Find the particular solution given that (i) $y(1) = \frac{1}{2}$, (ii) $y(2) = 0$.

SOLUTION

a First, the constant function $y = 0$ is trivially a solution. Now, because $y \neq 0$, we can write the DE as $-y^{-2}\,dy = x\,dx$, and integrating,

$$\int -y^{-2}\,dy = \int x\,dx, \qquad \text{so for some constant } C, \quad y^{-1} = \tfrac{1}{2}x^2 + C. \qquad (*)$$

b (i) Substituting $y(1) = \frac{1}{2}$ into $(*)$: $2 = \frac{1}{2} + C$, so $C = \frac{3}{2}$ and $y^{-1} = \frac{1}{2}x^2 + \frac{3}{2}$, that is $y = \dfrac{2}{x^2+3}$.

(ii) Substituting $y(2) = 0$ into $(*)$ is impossible because of division by zero, but the first solution $y = 0$ satisfies $y(2) = 0$, so it is the required solution.

CONCEPT 3 · dealing with absolute values

WHEN THE INTEGRAL IS A LOGARITHM

Many solutions involve absolute values, because $\dfrac{1}{y}$ has primitive $\log|y|$. Although the constant C may take any value, the power e^C is never negative and never zero. When interpreting the absolute value signs:

- Acknowledge that e^C is always positive.
- Modify the solution when removing the absolute value signs (the constant can then be positive or negative).
- Modify the solution again if any constant function is a solution (often this allows the constant to be zero too).

TUTOR DEMO **Absolute values**

a Solve $y' = 2x(3 - y)$. **b** Find the particular solution through (i) the origin, (ii) $(1, 3)$.

YOUR TURN **Same method**

a Solve $y' = y \cos x$. **b** Find the solution for which $y(0) = 2$.

WORKED EXAMPLE 3 **Keep for homework**

a Solve the DE $y' = x(1 - y)$. **b** Find the particular solution passing through (i) the origin, (ii) $(1, 1)$.

SOLUTION

a First, the constant function $y = 1$ is trivially a solution. Now, because $y \neq 1$, we can write the DE as $\dfrac{dy}{1-y} = x\,dx$, and integrating,

$$\int \frac{dy}{y-1} = -\int x\,dx$$

$$\log|y-1| = -\tfrac{1}{2}x^2 + C, \quad \text{for some constant } C$$

$$|y-1| = e^{-\frac{1}{2}x^2 + C}$$

$$|y-1| = Ae^{-\frac{1}{2}x^2}, \quad \text{where } A = e^C \text{ is positive.}$$

Hence $y - 1 = Ae^{-\frac{1}{2}x^2}$, where A can be positive or negative. We have already remarked that $y - 1 = 0$ is a solution of the original DE, so the general solution is $y = 1 + Ae^{-\frac{1}{2}x^2}$, for any constant A.

b (i) Substituting $(0, 0)$ gives $0 = 1 + A$, so $A = -1$ and $y = 1 - e^{-\frac{1}{2}x^2}$. (ii) Substituting $(1, 1)$ gives $1 = 1 + Ae^{-\frac{1}{2}}$, so $A = 0$, giving the constant function $y = 1$.

TEXTBOOK HOMEWORK · Exercise 13C

FOUNDATION

1 Consider the differential equation $\dfrac{dy}{dx} = \dfrac{x-1}{y+1}$.

a Multiply through by $y+1$ and then by dx, so that the variables are separated.

b Use the result $\displaystyle\int (x+a)^n\,dx = \frac{1}{n+1}(x+a)^{n+1} + C$ to find the general solution of this differential equation. Write your answer without using fractions.

2 Likewise find the general solutions of these separable equations. Make y the subject of the solution in each case.

a $\dfrac{dy}{dx} = xe^{-y}$

b $\dfrac{dy}{dx} = 4x^3(1+y^2)$

3

a Explain why $y = 0$, where $x \neq 0$, is a solution of $\dfrac{dy}{dx} = -\dfrac{y^2}{x}$. (Always look first for constant solutions.)

b Use the method of separable DEs to find the other solutions.

4 Suppose that the solution curve for $\dfrac{dy}{dx} = -\dfrac{x}{y}$ passes through $(1, \sqrt{3})$.

a Separate the variables and hence write down the corresponding equation of integrals.

b The general solution of this DE is a relation, not a function. Find the general solution, writing your answer without fractions.

c Hence determine the equation of the solution curve through the given point.

5 Likewise, for each DE, find the solution curve passing through the given point.

a $\dfrac{dy}{dx} = \dfrac{x}{y}$, through $(0, 1)$

b $\dfrac{dy}{dx} = (1 + x)(1 + y^2)$, with $y(-1) = 0$

c $\dfrac{dy}{dx} = -2y^2x$, with $y(1) = \frac{1}{2}$

d $\dfrac{dy}{dx} = e^{-y}\sec^2 x$, through $\left(\frac{\pi}{4}, \log 2\right)$

DEVELOPMENT

6 Consider the differential equation $\dfrac{dy}{dx} = \dfrac{2y + 4}{x}$.

a Find the constant solution, substituting to show that it is a solution of the DE.

b Use separation of variables to find the other solutions of the DE.

c How can the solutions in parts a and b be combined?

7 Consider the differential equation $y' = -xy$.

a Find the constant solution, substituting to show it is a solution of the DE.

b Use separation of variables to find the other solutions of the DE.

c How can the solutions in parts a and b be combined?

8 Use a similar approach to Questions 6 and 7 to solve these DEs.

a $\dfrac{dy}{dx} = \dfrac{2-y}{x}$ b $\dfrac{dy}{dx} = \dfrac{xy}{1+x^2}$ c $\dfrac{dy}{dx} = \dfrac{-2y}{x}$

d $\dfrac{dy}{dx} = y\sin x$ e $\dfrac{dy}{dx} = \dfrac{3y}{x^2}$ f $\dfrac{dy}{dx} = \dfrac{y(1-x)}{x}$

9

a Find all the constant solutions of $\dfrac{dy}{dx} = 3x^2\cos^2 y$ in the interval $-2\pi \leq y \leq 2\pi$.

b Find all the non-constant solutions.

10 Consider the initial value problem $\dfrac{dy}{dx} = \dfrac{2y}{x-1}$ with $y(2) = 1$.

a Show that the constant solution of the DE is not a solution of the IVP.

b Use separation of variables to find the general solution of the DE.

c Hence solve the IVP.

11 Consider the initial value problem $\dfrac{dy}{dx} = (y-1)\tan x$ with $y\left(\frac{\pi}{4}\right) = 3$.

a Show that the constant solution of the DE is not a solution of the IVP.

b Use separation of variables to find the general solution of the DE.

c Hence solve the IVP.

12 Use a similar approach to Questions 10 and 11 to solve these IVPs.

a $\dfrac{dy}{dx} = \dfrac{y}{x}$ with $y(2) = 1$

b $\dfrac{dy}{dx} = \dfrac{y}{2x}$ with $y(1) = 2$

c $\dfrac{dy}{dx} = \dfrac{-2xy}{1+x^2}$ with $y(1) = 2$

d $\dfrac{dy}{dx} = -\dfrac{y}{x}$ with $y(2) = 1$

e $\dfrac{dy}{dx} = y\cos x$ with $y\left(\frac{\pi}{2}\right) = 1$

f $\dfrac{dy}{dx} = \dfrac{y(2-x)}{x^2}$ with $y(2) = \frac{1}{2}$

13 a Differentiate $\log(\log x)$.

b Hence find the general solution of $(x \log x)\, y' = y$.

14 a Show that $\dfrac{x}{x+2} = 1 - \dfrac{2}{x+2}$.

b Hence solve the initial value problem $(x+2)\,y' - xy = 0$ with $y(0) = 1$.

15 a Use the double-angle formulae to rewrite $2\cos^2 x$ in terms of $\cos 2x$.

b The solution of the DE $\dfrac{dy}{dx} = \dfrac{2\cos^2 x}{y}$ is a relation and not a function. Use part a to find its equation, given that it passes through $(0, \sqrt{2})$.

16 a Let $y = x \times u$, where u is an unknown function of x. Use the product rule to find an expression for y'.

b Consider the differential equation $xy' = 2x + 2y$. (i) Use the result of part a to write a corresponding differential equation for u that is separable. (ii) Solve this DE for u. (iii) Hence write down the general solution of $xy' = 2x + 2y$.

ENRICHMENT

17 Suppose that $(x^2+1)\,y' + (y^2+1) = 0$ with $y(0) = 1$.

a Find a general solution of this DE.

b Show that the general solution is equivalent to $\dfrac{y+x}{1-xy} = D$.

c Hence find the solution of the IVP. Make y the subject of your answer.

PAST-PAPER HOMEWORK · mini exam

Suggested time: 30 minutes **19 marks** Calculator allowed · show working · hints are below each question if you get stuck

PP 1 **Sydney Grammar School, 2020, Trial Examination, Form VI Mathematics Extension 1, Q12(b)** 3 marks

Solve the initial value problem where $\dfrac{dy}{dx} = 2xe^{-y}$ given $y(0) = \ln 4$. Express your answer with y as the subject.

Hint: multiply both sides by e^{y} before separating.

PP 2 **Abbotsleigh, 2021, Higher School Certificate Assessment Task 4, Mathematics Extension 1, Q11(f)** 3 marks

Find the solution curve with $y(0) = 2$ for the differential equation $\dfrac{dy}{dx} = \dfrac{y}{1+x^2}$.

Hint: $\displaystyle\int \frac{dy}{y} = \int \frac{dx}{1+x^2}$. Deal with the absolute value using the initial condition.

PP 3 **Hornsby Girls High School, 2023, Year 12 Higher School Certificate Trial Examination, Mathematics Extension 1, Q13(b)** 3 marks

Find the equation of the curve which satisfies the differential equation $\dfrac{dy}{dx} = \dfrac{1-x}{y}$ and passes through the point $(2, 0)$.

Hint: the answer is a relation. Complete the square to recognise it.

PP 4 **Sydney Grammar School, 2022, Trial Examination, Form VI Mathematics Extension 1, Q13(e)** 3 marks

Solve the separable differential equation $\dfrac{dy}{dx} = x(y^2+4)$ if the solution passes through the point $(0, 2)$. Express your solution as a function of x.

Hint: $\displaystyle\int \frac{dy}{y^2+4}$ is a $\tan^{-1}$ standard form.

PP 5 **Sydney Technical High School, 2023, Trial HSC, Extension 1 Mathematics, Q13(a)** 3 marks

Solve the differential equation $\dfrac{dy}{dx} = 3(y+2)\sqrt{x}$, given that $y(0) = 7$. Give your answer in the form $y = f(x)$.

Hint: $\displaystyle\int \frac{dy}{y+2} = \int 3x^{\frac{1}{2}}\,dx$.

PP 6 **Blacktown Boys' High School, 2020, HSC Trial Examination, Mathematics Extension 1, Q13(d)** 4 marks

Solve the differential equation $\dfrac{dy}{dx} = \dfrac{1}{(2-y)\sqrt{2-x^2}}$ given that when $x = 1$, $y = 0$. Express y as a function of x.

Hint: you will get a quadratic in y. Use the initial condition to choose the correct sign of the square root.

LESSON 13D

y′ = g(y) and the logistic equation

Solve autonomous DEs $y' = g(y)$ by separating or by taking reciprocals, including exponential growth $y' = ky$ and the logistic equation $y' = ky(P - y)$, and use the DE itself to analyse gradient and concavity.

CONCEPT 1 · solving y′ = g(y)

TWO EQUIVALENT APPROACHES

Solve $y' = g(y)$ by integration using either of the forms

$$\frac{1}{g(y)}\,dy = dx \qquad \text{OR} \qquad \frac{dx}{dy} = \frac{1}{g(y)}.$$

A DE that does not involve the independent variable x is called **autonomous**. Laws of physics are usually independent of time, so autonomous DEs model populations, radioactive decay and cooling.

TUTOR DEMO **Both approaches**

Solve $y' = e^{2y}$ using both approaches, then make y the subject.

WORKED EXAMPLE 1 **Keep for homework**

Solve $y' = e^{y}$ using both approaches.

SOLUTION

EITHER regard it as a separable DE, $e^{-y}\,dy = dx$, and integrate:

$$-\int -e^{-y}\,dy = -\int dx$$
$$e^{-y} = -x + C$$
$$x = -e^{-y} + C$$

OR take reciprocals, $\dfrac{dx}{dy} = e^{-y}$, and integrate:

$$x = -e^{-y} + C, \quad \text{for some constant } C.$$

After either approach, solving for y gives $y = -\log(-x + C)$.

CONCEPT 2 · the exponential growth DE

Y′ = KY

Exponential growth has the DE $y' = ky$ ($k \neq 0$): the rate of change of the quantity is proportional to the quantity. Growth occurs when k is positive, and decay when k is negative. Its general solution is $y = Ae^{kx}$, where A can be any real number. (The constant function $y = 0$ is the case $A = 0$.)

TUTOR DEMO **An IVP**

Solve $y' = 2y$ given the initial value $y(1) = 10$.

YOUR TURN **Shifted constant**

Solve $y' = 4 - y$, with $y(0) = 1$. (Find the constant solution first.)

WORKED EXAMPLE 2 **Keep for homework**

Solve the differential equation $y' = -3y$ given the initial value $y(2) = 50$.

SOLUTION

First, the constant function $y = 0$ is trivially a solution. Otherwise, rearranging, $\frac{1}{y}\,dy = -3\,dx$, and integrating,

$$\log|y| = -3x + C, \quad \text{for some constant } C$$
$$|y| = e^{-3x+C} = Ae^{-3x}, \quad \text{where } A = e^C$$
$$y = Ae^{-3x}, \quad \text{where } A \text{ can be positive or negative.}$$

Because the constant function $y = 0$ is a solution, the general solution is $y = Ae^{-3x}$, where A can be any real number. Substituting $y(2) = 50$ gives $50 = Ae^{-6}$, so $A = 50e^6$, and $y = 50e^{-3(x-2)}$.

CONCEPT 3 · the logistic equation

THE LOGISTIC DIFFERENTIAL EQUATION

- In this course a logistic DE has the form $y' = ky(P - y)$, where k and P are constants (assumed positive below).
- The constant functions $y = 0$ and $y = P$ are solutions.
- Otherwise there are three groups of solution curves: between $y = 0$ and $y = P$, with domain all real x; above $y = P$; and below $y = 0$.
- Solving needs a **partial fractions** decomposition, which will always be given in a question.
- In applications, one group, or even two groups, may have no significance.

WORKED EXAMPLE 3 **Keep for homework**

a Show that $\dfrac{1}{y(1-y)} = \dfrac{1}{y} + \dfrac{1}{1-y}$. **b** Hence solve $y' = y(1-y)$, writing y as the subject.

SOLUTION

a RHS $= \dfrac{(1-y)+y}{y(1-y)} =$ LHS.

b First, $y = 0$ and $y = 1$ are trivially solutions. Otherwise, using part a,

$$\left(\frac{1}{y} + \frac{1}{1-y}\right)dy = dx$$

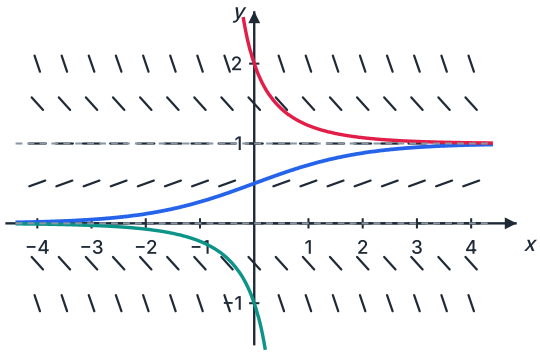
$$\log|y| - \log|1-y| = x + C$$

$$\left|\frac{y}{1-y}\right| = Ae^x, \quad A = e^C > 0$$

$$\frac{y}{1-y} = Ae^x, \quad A \neq 0$$

$$y = \frac{Ae^x}{Ae^x + 1} = \frac{1}{1 + Be^{-x}}$$

where $B = A^{-1}$. Hence $y = 0$, or $y = 1$, or $y = \dfrac{1}{1+Be^{-x}}$ for some non-zero constant B.



Solution curves of $y' = y(1-y)$ through $\left(0, \frac{1}{2}\right)$ (blue, $B = 1$), $(0, 2)$ ($B = -\frac{1}{2}$) and $(0, -1)$ ($B = -2$). No solution crosses $y = 0$ or $y = 1$.

TUTOR DEMO **Picking a solution curve**

Using the general solution $y = \dfrac{1}{1 + Be^{-x}}$ of $y' = y(1 - y)$, find the solution curve through $\left(0, \frac{1}{2}\right)$. State its domain, range and asymptotes, and describe it if x is time and y is population.

YOUR TURN **A different region**

Find the solution curve of $y' = y(1 - y)$ through $(0, 2)$. Find its vertical asymptote, and the domain and range of the connected curve through $(0, 2)$.

CONCEPT 4 · using the DE to find the second derivative

CONCAVITY STRAIGHT FROM THE DE

Rather than differentiating the solution twice, differentiate the DE itself with the chain rule:

$$y'' = \frac{d}{dy}\big(g(y)\big) \times \frac{dy}{dx} = g'(y)\, y'.$$

WORKED EXAMPLE 4 **Keep for homework**

For $y' = y(1-y)$: **a** prove that $y'' = y(1-y)(1-2y)$; **b** use this and the DE to analyse the gradient and concavity of the solution curves.

SOLUTION

a Expanding, $y' = y - y^2$. Using the chain rule,

$$y'' = \frac{d}{dy}(y - y^2) \times \frac{dy}{dx} = (1-2y)\,y' = y(1-y)(1-2y).$$

b y' is positive for $0 < y < 1$ and negative for $y < 0$ or $y > 1$, so curves in the middle region always increase and those in the top and bottom regions always decrease. y'' is positive for $0 < y < \frac{1}{2}$ or $y > 1$, and negative for $\frac{1}{2} < y < 1$ or $y < 0$. So the top region is always concave up, the bottom region always concave down, and curves in the middle region change from concave up to concave down at $y = \frac{1}{2}$, giving a point of inflection there.

YOUR TURN **Same method**

For $y' = 2y(3-y)$, find y'' in terms of y alone, and hence find the height y at which the solution curves between $y = 0$ and $y = 3$ have a point of inflection.

TEXTBOOK HOMEWORK · Exercise 13D

FOUNDATION

1 Consider the differential equation $\frac{dy}{dx} = -y$.

a What is the constant solution of this equation?

b Use separation of variables to solve the DE.

c Make y the subject of this solution, simplifying the constant part of the expression.

d Check that the constant solution is included in your answer to part c.

e Find the solution for the initial condition $y(0) = 2$.

2 Consider the differential equation $\frac{dy}{dx} = 3y$.

a What is the constant solution of this equation?

b Write down the DE obtained by taking the reciprocal of both sides.

c Use direct integration to obtain x as a function of y.

d Make y the subject of this solution, simplifying the constant part of the expression.

e Check that the constant solution is included in your answer to part d.

f Find the solution for the initial condition $y(0) = -1$.

3 Find the solutions of these autonomous IVPs (autonomous means that it has the form $y' = g(y)$). Use either of the methods given in Questions 1 and 2.

a $y' - y = 0$, with $y(0) = -3$

b $y' + 2y = 0$, with $y(0) = 1$

c $y' = -3y$, with $y(0) = 2$

d $y' = 2y$, with $y(0) = -1$

4 Consider the differential equation $\frac{dy}{dx} = 2 - y$.

a What is the constant solution of this equation?

b Use separation of variables or take reciprocals to solve this DE.

c Make y the subject, simplifying the constant part of the expression.

d Check that the constant solution is included in your answer to part c.

e Find the solution for the initial condition $y(0) = 3$.

5 Follow the procedures in Question 4 to solve these IVPs.

a $y' = 1 - y$, with $y(0) = 3$

b $y' = y - 1$, with $y(0) = 0$

c $y' = \frac{1}{2}(y + 1)$, with $y(0) = 1$

d $y' = 2(3 - y)$, with $y(0) = 4$

6 Solve these initial value problems.

a $y' = 2y^2$, with $y(0) = 3$

b $y' = -y^2$, with $y(0) = 1$

c $y' = 1 + y^2$, with $y\left(\frac{\pi}{4}\right) = 1$

d $y' = -e^y$, with $y(0) = 0$

DEVELOPMENT

7 Consider the differential equation $y' = ky$, where k is an unknown constant.

a Find the general solution of this DE.

b Evaluate the arbitrary constant, given the initial condition $y(0) = 20$.

c Finally, evaluate k given the condition $y(2) = 5$.

d Simplify your solution, and hence evaluate $y(3)$.

8 Once again consider the differential equation $y' = ky$, where k is an unknown constant.

a Find the general solution of this DE.

b Evaluate the arbitrary constant, given the initial condition $y(0) = 8$.

c Finally, evaluate k given the condition $y(2) = 18$.

d Simplify your solution, and hence evaluate $y(4)$.

9 a Find the general solution of the DE $y' = e^{-y}$.

b Describe the family of curves that you have found.

c Draw the slope field for this DE. (i) Draw the solution curve that passes through the origin. (ii) Draw two other solution curves. (iii) Describe how your two solution curves can be obtained by simple transformations. (iv) What feature of the slope field makes this possible?

d Evaluate the arbitrary constant, given that the curve passes through $(0, 1)$.

10 Let $L(x) = \dfrac{1}{1+e^{-x}}$ and consider the curve $y = L(x)$.

a What is the y-intercept?

b Explain why the function is always positive.

c Determine $\lim\limits_{x\to\infty} L(x)$ and $\lim\limits_{x\to-\infty} L(x)$.

d Find L' and hence show that the curve has no stationary points.

e (i) Show that $L' = \dfrac{1}{\left(e^{\frac{x}{2}} + e^{-\frac{x}{2}}\right)^2}$. (ii) Use the result in part i to find L''. (iii) Hence find the point of inflection of $y = L(x)$.

f Sketch $y = L(x)$.

g Show by substitution that $y = L(x)$ is a solution of the logistic DE $y' = y(1-y)$.

11 **a** Show that $\dfrac{1}{y(1-y)} = \dfrac{1}{y} + \dfrac{1}{1-y}$.

b Consider the logistic differential equation $y' = y(1-y)$. (i) What are the constant solutions of this equation? (ii) Use part a to find the general solution of the logistic DE.

c Show that the solution is the result of shifting the function in Question 10, and determine the shift.

d The constant solutions cannot be obtained in the usual way from the general solution. Use the general solution $y = \dfrac{1}{1+Be^{-x}}$ to answer: (i) Find $\lim\limits_{B\to\infty} y$. Is this one of the constant solutions? (ii) Find $\lim\limits_{B\to 0^+} y$. Is this one of the constant solutions?

12 A more generalised version of the logistic equation is $y' = ry(1-y)$, for some constant r.

a Find the constant solutions.

b Use part a of Question 11 to find the general solution.

c Suppose that the initial condition is $y(0) = y_0$. Determine the value of the arbitrary constant B given in the answer to part b.

d By using the answer to part c, or otherwise, what happens as: (i) $y_0 \to 0^+$, (ii) $y_0 \to 1^-$?

13 **a** Show that $\dfrac{1}{(y-1)(y-3)} = \dfrac{1}{2}\left(\dfrac{1}{y-3} - \dfrac{1}{y-1}\right)$.

b Consider the modified logistic equation $y' = -(1-y)(3-y)$. (i) What are the constant solutions? (ii) Use part a to find the general solution of the given DE.

c Using the general solution given in the answers: (i) Which of the constant solutions is found by taking $B \to 0^+$? (ii) Which of the constant solutions is found by taking $B \to \infty$?

d (i) Find an expression for y'' in terms of y alone. (ii) Hence determine the location of the inflection point for the solution curve with $B = 1$.

ENRICHMENT

14 Care must be taken when the solution of a DE involves inverse functions. Consider the initial value problem $\frac{dy}{dx} = -\sqrt{1-y^2}$, with $y(0) = 1$.

a Find the implicit general solution of the DE.

b Evaluate the unknown constant by applying the initial condition.

c Making y the subject without taking care, it would seem that the solution of the IVP is $y = \cos x$. Explain why this solution is not valid for all values of x. It may help to substitute this solution into each side of the DE.

d What is the correct solution of the IVP in which y is the subject?

PAST-PAPER HOMEWORK · mini exam

Suggested time: 30 minutes **21 marks** Calculator allowed · show working · hints are below each question if you get stuck

PP 1 Kinross Wolaroi School, 2023, Year 12 Mathematics Extension 1 Trial HSC Examination, Q8 — 1 mark

If $\dfrac{dN}{dt} = 0.1(N - 200)$ which of the following is a correct expression for N?

(A) $-200 - 100e^{0.1t}$ (B) $-100 - 200e^{0.1t}$ (C) $200 + 100e^{0.1t}$ (D) $100 + 200e^{0.1t}$

Hint: the general solution is $N - 200 = Ae^{0.1t}$.

PP 2 Hurlstone Agricultural High School, 2021, Higher School Certificate Assessment Task 4, Mathematics Extension 1, Q13(c) — 2 marks

Solve the differential equation $\dfrac{dy}{dx} = \cos^2 y$ given that $y = \dfrac{\pi}{4}$ when $x = 0$.

Hint: $\dfrac{1}{\cos^2 y} = \sec^2 y$.

PP 3 Gosford High School, 2023, Trial HSC Examination, Mathematics Extension 1, Q13(b) — 4 marks

Show that the solution to the differential equation $\dfrac{dy}{dx} = 1 + \dfrac{2y}{3}$, given $y = -1$ when $x = 2$, can be written in the form

$$y = \frac{e^{\frac{2(x-2)}{3}} - 3}{2}.$$

Hint: write the RHS as $\dfrac{3 + 2y}{3}$, then separate. Decide the sign of $3 + 2y$ from the initial condition.

PP 4 PLC, 2023, HSC Trial Examination, Mathematics Extension 1, Q14(a) — 5 marks

An area A cm² is occupied by a bacteria colony. The colony has a growth rate modelled by the logistic equation

$$\frac{dA}{dt} = \frac{1}{25}A(50 - A) \quad \textbf{where } t \geq 0 \textbf{ and is measured in days.}$$

At time $t = 0$, the area occupied by the bacteria colony is $\frac{1}{2}$ cm².

(i) Show that $\dfrac{1}{A(50-A)} = \dfrac{1}{50}\left(\dfrac{1}{A} + \dfrac{1}{50-A}\right)$. 1

(ii) Solve the logistic equation and hence show that $A = \dfrac{50}{1 + 99e^{-2t}}$. 3

(iii) What is the limiting area of the bacteria colony? 1

Hint: after integrating you get $\frac{1}{50}\log\left|\frac{A}{50-A}\right| = \frac{t}{25} + C$. Note that $0 < A < 50$ throughout.

PP 5 Baulkham Hills High School, 2023, Higher School Certificate Trial Examination, Mathematics Extension 1, Q14(b) — 4 marks

In the town of Eulerville, the population was estimated in the year 2000 to be 50 000. The population growth is given by the logistic equation $\dfrac{dP}{dt} = 0.05P\left(\dfrac{C-P}{C}\right)$, where t is the number of years after 2000 and C is the carrying capacity.

In the year 2023, the population was estimated to be 80 000. Use the fact that $\dfrac{C}{P(C-P)} = \dfrac{1}{P} + \dfrac{1}{C-P}$ to find the carrying capacity, to the nearest thousand.

Hint: solve to get $P = \dfrac{C}{1 + Be^{-0.05t}}$, use $t = 0$ to write B in terms of C, then solve for C using $t = 23$.

PP 6 **Gosford High School, 2023, Trial HSC Examination, Mathematics Extension 1, Q13(c)** 5 marks

A rabbit population grows according to the logistic equation $\dfrac{dP}{dt} = 0.2P\left(1 - \dfrac{P}{10000}\right)$, where P is the number of rabbits after t months. The initial population is 1000.

(i) Show that $\dfrac{10000}{P(10000-P)} = \dfrac{1}{P} + \dfrac{1}{10000-P}$. 1

(ii) Find the population of rabbits after seven months. 4

Hint: rewrite the DE as $\dfrac{dP}{dt} = \dfrac{0.2}{10000}P(10000-P)$ before separating.

LESSON 13E

Applications of differential equations

Find unknown constants when the form of the solution is known, then model real situations with the natural growth DE, Newton's law of cooling and the logistic equation. The variable is usually t for time.

CONCEPT 1 · when the form of the solution is known

EQUATING COEFFICIENTS

People who work with DEs learn to recognise them, so they often know what the solution looks like but not the values of the constants. Substitute the given form into the DE, then equate coefficients (of x, of each exponential, and the constants) to find them.

TUTOR DEMO Equate coefficients

a Find the values of a and b so that $y = ax + b + 2e^{-x}$ is a solution of $y' + y = 4x$. **b** Find n so that $y = 3\sin nx$ is a solution of $y'' = -16y$.

WORKED EXAMPLE 1 Keep for homework

a Find the values of b and c so that $y = e^{3x} + bx + c$ is a solution of $y' = 3y - 18x$. **b** Find n so that $y = 10\cos nx$ is a solution of the second-order DE $y'' = -49y$.

SOLUTION

a Substituting $y = e^{3x} + bx + c$ into the DE,

$$\text{LHS} = 3e^{3x} + b \qquad \text{RHS} = 3e^{3x} + (3b - 18)x + 3c$$

Equating coefficients of x: $3b - 18 = 0$. Equating constants: $3c = b$. So $b = 6$ and $c = 2$, giving $y = e^{3x} + 6x + 2$.

b Substituting $y = 10\cos nx$: LHS $= -10n^2\cos nx$, RHS $= -490\cos nx$. Hence $-10n^2 = -490$, so $n = 7$ or $n = -7$, giving $y = 10\cos 7x$ and $y = 10\cos(-7x)$, which are the same function because cosine is even.

CONCEPT 2 · natural growth in a practical situation

FORMING AND SOLVING THE DE

1. Translate the words into a DE: "the rate of increase of N is proportional to N" means $\frac{dN}{dt} = kN$.
2. Solve the DE, remembering any constant solution and the absolute values.
3. Use the two given data points to find the arbitrary constant and k.

The same steps handle "the rate of change of temperature is proportional to the difference from the surroundings" (Newton's law of cooling, $\frac{dT}{dt} = k(T - R)$).

TUTOR DEMO **Newton's law of cooling**

A cup of tea at 90°C is left in a room at 20°C, and its temperature T obeys $\frac{dT}{dt} = k(T - 20)$. After 10 minutes it is 60°C. **a** Solve the DE and find k. **b** When will the tea reach 40°C?

YOUR TURN **Radioactive decay**

A radioactive mass M grams decays according to $\frac{dM}{dt} = kM$. Initially there are 80 grams, and after 5 days there are 60 grams. Find M as a function of t, and the half-life correct to the nearest tenth of a day.

WORKED EXAMPLE 2 **Keep for homework**

The rabbits on Bandicoot Island are increasing. Fifty years ago, 100 rabbits were released, and now there are 10 000 rabbits. Assume that the rate of increase of rabbits is proportional to the number of rabbits. **a** Find the equation for the population N at time t years after they were introduced. **b** Find how many rabbits there will be in another 25 years.

SOLUTION

a As a DE, $\dfrac{dN}{dt} = kN$. First, $N = 0$ is trivially a solution. Otherwise $\dfrac{1}{N}\,dN = k\,dt$, and integrating, $\log|N| = kt + C$, so $N = Ae^{kt}$, where A can be any real number.

When $t = 0$, $N = 100$, so $A = 100$ and $N = 100e^{kt}$. When $t = 50$, $N = 10\,000$, so $10\,000 = 100e^{50k}$ and $k = \frac{1}{50}\log 100$.

b Substituting $t = 75$: $kt = \frac{1}{50}\log 100 \times 75 = 1.5\log 100$, so $N = 100e^{1.5\log 100} = 100 \times 100^{1.5} = 100\,000$ rabbits.

CONCEPT 3 · the logistic equation as a limit to growth

LOGISTIC GROWTH

Populations cannot grow exponentially forever: food is limited and there may be predators. The simplest model of a limit is the logistic equation $\dfrac{dN}{dt} = kN(P - N)$, where P is the **carrying capacity** (maximum population). Its solution between 0 and P is

$$N = \frac{P}{1 + Be^{-Pkt}},$$

found using the partial fractions $\dfrac{1}{N(P-N)} = \dfrac{1}{P}\left(\dfrac{1}{N} + \dfrac{1}{P-N}\right)$. The population rises rapidly, then levels off towards P.

WORKED EXAMPLE 3 **Keep for homework**

The carrying capacity of Bandicoot Island is 20 000 rabbits. Using the data of Worked Example 2 (100 rabbits at $t = 0$, 10 000 at $t = 50$), model the population by $\frac{dN}{dt} = kN(P - N)$ with $P = 20\,000$. Find N, and the population in another 25 years.

SOLUTION

First, $N = 0$ and $N = P$ are trivially solutions. Otherwise, using partial fractions,

$$\left(\frac{1}{N} + \frac{1}{P - N}\right) dN = Pk\,dt$$

$$\log|N| - \log|P - N| = Pkt + C$$

$$\frac{N}{P - N} = Ae^{Pkt} \quad (A \text{ positive or negative})$$

$$N = \frac{P}{1 + Be^{-Pkt}} \quad (B = A^{-1})$$

When $t = 0$, $N = 100$: $100 = \frac{20\,000}{1 + B}$, so $B = 199$. When $t = 50$, $N = 10\,000$: $1 + 199e^{-1\,000\,000k} = 2$, so $k = \frac{\log 199}{1\,000\,000}$.

When $t = 75$: $-Pkt = -1.5 \log 199$, so $N = \frac{20\,000}{1 + 199e^{-1.5 \log 199}} \approx 18\,676$ rabbits, far fewer than the 100 000 predicted by exponential growth.

YOUR TURN **Logistic model**

A population obeys $\frac{dN}{dt} = kN(500 - N)$, with $N(0) = 50$ and $N(2) = 100$. Find N as a function of t, and the population when $t = 6$.

TEXTBOOK HOMEWORK · Exercise 13E

FOUNDATION

1 a In each case find the values of a and b given that: (i) $y = ax^2 + bx$ is a solution of $y' = 1 - 4x$; (ii) $y = e^{-x}(a\cos x + b\sin x)$ is a solution of $y' = 2e^{-x}\sin x$; (iii) $y = ax + b + 3e^{-x}$ is a solution of $y' = x - y$.

b In each case find the values of a, b and c given that: (i) $y = ax^2 + bx + c + 4e^{-2x}$ is a solution of $y' + 2y = x^2 - 3x - 4$; (ii) $y = ax^2 + bx + c - \sin 2x$ is a solution of $y'' + 4y = x^2 + 5x$.

c Find the possible values of λ given that $y = 5e^{\lambda x}$ is a solution of $y'' + 5y' + 6y = 0$.

2 In a laboratory, a scientist has a sample of radioactive material. The material decays at a rate proportional to the amount present, that is $\frac{dR}{dt} = kR$, where R is the amount present at time t days, and k is an unknown constant.

a Find the general solution of this DE.

b Initially there is 100 grams of the material. Determine the arbitrary constant in your solution.

c After 4 days only 20 grams of the substance remains radioactive. Determine the value of k.

d Hence determine the amount present after 12 days.

3 A metal ingot is put in a fridge until its temperature is 5°C. The ingot is then taken out of the fridge and left in a room maintained at 25°C. Let H be the temperature of the ingot after t minutes. Experiments show that $\frac{dH}{dt} = k(H - 25)$, for some constant k.

a Find the general solution of this DE.

b Use the initial condition to determine the arbitrary constant.

c After 10 minutes the ingot is at 15°C. Determine the value of k.

d Find how long it takes, correct to the nearest minute, for the temperature to reach 24°C.

4 A certain tank is in the shape of a cylinder. It is filled with water to a height of 400 cm. A tap at the bottom of the tank is opened, and as the water empties, the rate of change of height of water in the tank is proportional to the square root of the height: $\frac{dh}{dt} = k\sqrt{h}$.

a State the initial condition, and explain why the constant k must be negative.

b Solve the IVP. You may assume that the arbitrary constant of integration is positive.

c After 20 minutes the height of the water in the tank is 100 cm. Find the value of k.

d How long does the tank take to drain?

e Is the function you found in part b valid for larger values of t?

5 A certain curve has the property that the tangent at any point passes through the origin.

a Write down the gradient of the line from $(0, 0)$ to the point (x, y).

b Now suppose that (x, y) is on this curve. Write down a differential equation for this curve.

c Hence determine the general equation of this curve.

d Which special case is a solution of the problem, but not of the DE?

6 A certain curve has the property that the normal at any point passes through the origin.

a Write down the gradient of the line from $(0, 0)$ to the point (x, y).

b Now suppose that (x, y) is on this curve. Write down a differential equation for this curve.

c Hence determine the general equation of this curve, which is a relation and not a function.

7 A tangent is drawn to a curve, and it is found that its x-intercept is 1 less than the x-coordinate of the point of contact.

a Write down the gradient of the line from $(x - 1, 0)$ to the point (x, y).

b Now suppose that (x, y) is on this curve. Write down a differential equation for this curve.

c Hence determine the general equation of this curve.

d Find such a curve passing through $(0, 1)$.

DEVELOPMENT

8 The atmospheric pressure P on the planet Nebula changes with altitude h at a rate proportional to P, that is $\frac{dP}{dh} = kP$, for some constant k. Let the atmospheric pressure at ground level be $P = P_0$.

a Find the general solution of this DE.

b Measurements from satellites orbiting the planet show that the pressure at 10 000 m is 40 kPa, and the pressure at 6000 m is 80 kPa. Find the value of k.

c Determine the pressure at ground level correct to the nearest kPa.

9 A tangent to a curve intersects the coordinate axes at A and B. It is found that the point of contact of the tangent is also the midpoint of AB.

a Let the point of contact with the curve be (x, y). Find the coordinates of A and B in terms of x and y.

b Use the gradient of AB to determine a differential equation for this curve.

c Hence determine the general equation of this curve.

10 According to Fick's law, diffusion across a cell membrane is governed by a differential equation. If $C(t)$ is the concentration of a solute in a cell, and S is the concentration of the solute in the surrounding medium, then $\dfrac{dC}{dt} = k(S - C)$, for some constant k.

a Find the general solution of this DE.

b Suppose that initially $C(0) = C_0$, where $C_0 < S$. Solve the IVP.

11 In order to save an endangered species, it has been decided to release 40 animals on an island where there are no predators. The carrying capacity is 1000. It is assumed that the population N of these animals at time t years after release fits the logistic growth equation $\dfrac{dN}{dt} = kN(1000 - N)$, for some constant k.

a Show that $\dfrac{1000}{N(1000 - N)} = \dfrac{1}{N} + \dfrac{1}{1000 - N}$.

b Use the result of part a to find the general solution of the logistic growth equation.

c Determine the arbitrary constant by applying the initial condition $N(0) = 40$, then simplify the function.

d Given that the population of animals after 1 year was 80, find the value of k correct to four significant figures.

e What will the population be after 5 years, correct to the nearest whole number?

12 In the mid-1800s, Verhulst estimated the population growth of the United States of America using the logistic differential equation $\dot{N} = kN(P - N)$, for some positive constant k, where N is the population in millions, and t is the number of years after 1850.

a Show that $\dfrac{P}{N(P-N)} = \dfrac{1}{N} + \dfrac{1}{P-N}$.

b Use the result of part a to find the general solution of the logistic growth equation.

c It was estimated at the time that the carrying capacity of the USA was $P = 187.5$, and the population in 1850 was recorded as $N(0) = 23.2$, both in units of millions. Determine the arbitrary constant and simplify the function.

d The estimate used for k was $k = 1.6 \times 10^{-4}$, which predicted a population of 59.8 million in 1890. The actual population of the USA in 1890 was 63.0 million. Calculate a new value for k.

e Using the revised value of k, compare the predicted population for 1930 using this model with the actual population, which was 123.2 million.

f The population of the USA in 2018 was approximately 327 million. Comment on this value.

13 Biologists are modelling the population of an endangered species of fish in a river system. The indigenous population of the area are permitted to harvest 200 fish once a year in January as part of their culture. Data collected in December indicate that there are 500 fish in the river system. The data also suggest the model $\frac{dy}{dt} = -2 + \frac{1}{24}y(16 - y)$, where y is the population measured in hundreds at time t years after the next harvest.

a The term $\frac{1}{24}y(16 - y)$ on the right-hand side represents the familiar logistic growth model. What is the significance of the -2 in this equation?

b Show that the DE can be re-written as $\frac{dy}{dt} = -\frac{1}{24}(y - 4)(y - 12)$, and write down the initial condition assuming that the harvest goes ahead.

c Show that $\frac{24}{(y-4)(y-12)} = \frac{3}{y-12} - \frac{3}{y-4}$.

d Hence solve the IVP in part b.

e According to this model, the fish in the river will die out. When will that be?

f (i) If the most recent harvest had not occurred, what would the initial condition change to? (ii) It can be shown that the solution of the DE for this initial condition is $y = \frac{4\left(3 + 7e^{-\frac{1}{3}t}\right)}{1 + 7e^{-\frac{1}{3}t}}$. Investigate what happens for this solution over time. Comment on the result.

ENRICHMENT

14 An object is heated to 100°C and then placed in a room at 20°C. Let h be the temperature of the object after t minutes. The rate of change of temperature is proportional to its difference from the room temperature, $\frac{dh}{dt} = k(h - 20)$. After 10 minutes the temperature of the object is 80°C.

a Solve the initial value problem and find the value of the constant k.

b At the 10-minute mark, refrigeration equipment is turned on, which lowers the temperature in the room by 1°C/min. Assume that the rate of change of temperature continues to be proportional to the difference, with the same k. Let $H(t)$ be the temperature of the object t minutes after the refrigeration is turned on. (i) Write down the initial value problem for $H(t)$ in terms of k. (ii) Let $y = H - 20 + t$. What is the corresponding IVP for y? (iii) Find y, and hence determine $H(t)$.

PAST-PAPER HOMEWORK · mini exam

Suggested time: 35 minutes **22 marks** Calculator allowed · show working · hints are below each question if you get stuck

PP 1 **Abbotsleigh, 2020, Higher School Certificate Assessment 4 Trial Examination, Mathematics Extension 1, Q13(b)** 4 marks

Miriam takes a saucepan of soup from the stove and measures its temperature as 96°C. The air temperature in Miriam's kitchen is 21°C. Miriam knows that the temperature T of her soup at time t minutes after she has removed it from the stove can be modelled using the differential equation

$$\frac{dT}{dt} = -k(T - R),$$

where R is the air temperature of the kitchen and k is a positive constant.

(i) Show that $T = 21 + 75e^{-kt}$ is a solution of the differential equation. 1

(ii) Miriam finds the temperature of the soup 5 minutes after she removes it from the stove is 87°C. If she likes to start eating it when it reaches 70°C, find how much longer Miriam will need to wait until the soup is cool enough. 3

Hint: "how much longer" is measured from the 5-minute mark, not from when the soup left the stove.

PP 2 **Sydney Grammar School, 2023, Trial Examination, Form VI Mathematics Extension 1, Q12(b)** 4 marks

The current world population is approximately 8 billion people. Assuming that the 'carrying capacity' of the earth is 25 billion people, Anna models the population P after t years using the differential equation

$$\frac{dP}{dt} = k(25 - P).$$

(i) Show that $P = 25 - 17e^{-kt}$ satisfies the differential equation above. 1

(ii) Find k if the world population after 1 year is 8.072 billion people. Write your answer correct to 3 significant figures. 2

(iii) After how many years does Anna's model say the world population will reach 20 billion people? Round your answer to the nearest whole year. 1

Hint: in (ii), make e^{-k} the subject and take logs.

PP 3 **Killara High School, 2023, Year 12 Mathematics Extension 1 Trial Examination, Q14(a)** 4 marks

Steve weighs 80 kg and decides to go on a diet of 1600 calories per day. His body automatically uses 850 calories each day to maintain his basic bodily functions. He spends 15 cal/kg/day doing exercise. 1 kg of fat stores 10 000 calories and the process of storing energy in fat is assumed to be 100% efficient, such that all 10 000 calories must be used to remove a full 1 kg of weight from a person, or 5000 calories need to be used to remove 0.5 kg of weight.

(i) Show that the rate of change of Steve's weight (w) over time (t) is given by the equation 1

$$\frac{dw}{dt} = \frac{750 - 15w}{10\,000}.$$

(ii) Solve the differential equation to find an expression for Steve's weight. 2

(iii) If he continues this diet and the exercise regime, will his weight approach an equilibrium in the long term? 1

Hint: calories in minus calories used per day, divided by the calories per kilogram, gives kilograms per day.

PP 4 **James Ruse Agricultural High School, 2020, Year 12 Trial HSC Examination, Mathematics Extension 1, Q13(b)** 5 marks

A tank contains a saltwater solution consisting initially of 20 kg of salt dissolved into 10 L of water. Fresh water is being poured into the tank at a rate of 3 L/min and the solution (kept uniform by stirring) is flowing out at 2 L/min.

(i) Show that Q, the amount of salt (in kilograms), at time t (in minutes) satisfies the equation $\dfrac{dQ}{dt} = -\dfrac{2Q}{(10+t)}$. 2

(ii) Solve the differential equation given in (i) to find the amount of salt in the tank after 5 minutes. Answer in kilograms correct to 2 decimal places. 3

Hint: the volume increases by 1 L each minute, so at time t the volume is $10 + t$ litres, and the concentration is $\dfrac{Q}{10+t}$.

PP 5 **Ascham School, 2023, Year 12 Mathematics Extension 1 Trial Examination, Q13(d)** 5 marks

A dam can sustainably carry 10 million fish. The fish population N (in millions) in the dam follows the logistic differential equation $\dfrac{dN}{dt} = \dfrac{N(10-N)}{200}$, where t is the time in months since the initial fish count. A fish farming company was unaware of this carrying capacity and introduces 15 million fish. Prior to this, the dam did not have fish.

Find the number of fish remaining after 6 months. Answer to the nearest million fish. (You may use the identity $\dfrac{10}{N(10-N)} = \dfrac{1}{N} + \dfrac{1}{10-N}$.)

Hint: here $N > 10$, so $\left|\dfrac{N}{10-N}\right| = \dfrac{N}{N-10}$.

CHAPTER 13 REVIEW

Chapter review exercise

Mixed practice from the whole chapter.

1 In each part, state the order of the DE, whether or not it is linear, and how many arbitrary constants will appear in the general solution.

a $y' + xy = \cos x$ b $y'' + x^2y' - 3y = e^x \sin x$ c $y''' + y''y' = x - 2$

2 Show by substitution that the given function with arbitrary constant C is a general solution of the accompanying differential equation.

a $y = Cx^2e^x; \quad xy' = y(2 + x)$ b $y = \sqrt{x^2 + C}; \quad y'y = x$ c $y = \dfrac{1}{x^2 + C}; \quad y' = -2xy^2$

3 Consider the differential equation $\dfrac{dy}{dx} = -\frac{1}{2}y$.

a What is the constant solution of this equation?

b Write down the DE obtained by taking the reciprocal of both sides.

c Use direct integration to obtain x as a function of y.

d Make y the subject of this solution, simplifying the constant part of the expression.

e Check that the constant solution is included in your answer to part d.

f Find the solution for the initial condition $y(0) = 3$.

4 Verify that the given function is a general solution of the differential equation for all values of the constants A, B, C and D.

a $y = Ae^{-x} + Be^{-2x}; \quad y'' + 3y' + 2y = 0$

b $y = Ae^{-x}\cos 2x + Be^{-x}\sin 2x; \quad y'' + 2y' + 5y = 0$

5 Use separation of variables to solve each differential equation. Note that the solution of part b is a relation that is not a function.

a $\dfrac{dy}{dx} = \dfrac{-2xy}{1+x^2}$ b $\dfrac{dy}{dx} = \dfrac{1-x}{2+y}$ c $\dfrac{dy}{dx} = \dfrac{y(x-1)}{x}$

6 Solve each IVP by taking the reciprocal and using direct integration.

a $y' = \frac{1}{2}(1-y)$, with $y(0) = 2$ b $y' = \frac{1}{3}(5-y)$, with $y(0) = 2$

7 a Show that $\dfrac{1}{1-x^2} = \dfrac{1}{2}\left(\dfrac{1}{1+x} + \dfrac{1}{1-x}\right)$.

b Hence solve $\dfrac{dy}{dx} = \dfrac{y}{1-x^2}$ by separation of variables.

8 Use separation of variables to solve each initial value problem.

a $\frac{dy}{dx} = -\frac{y}{x}$, with $y(2) = 1$

b $\frac{dy}{dx} = (1 + 2x)e^{-y}$, with $y(1) = 0$

c $\frac{dy}{dx} = \frac{y^2}{\sqrt{x}}$, with $y(0) = -1$

9 a Show that $\frac{1}{y(1-y)} = \frac{1}{y} + \frac{1}{1-y}$.

b Consider the logistic differential equation $y' = y(1 - y)$. (i) What are the constant solutions of this equation? (ii) Use part a to find the general solution of the logistic DE. (iii) Hence find the solution that passes through $\left(0, \frac{1}{4}\right)$.

10 a Show that $\frac{1}{(2-y)(3-y)} = \frac{1}{y-3} - \frac{1}{y-2}$.

b Consider the harvest logistic differential equation $y' = -\frac{1}{5}(3 - y)(2 - y)$. (i) What are the constant solutions of this equation? (ii) Use part a to find the general solution of the harvest logistic DE with initial condition $y(0) = 1$. (iii) For what value of x does $y = 0$?